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Mobile Computing LAB

1. **Use Figma to create an interactive diagram illustrating the components of a basic cellular system.**

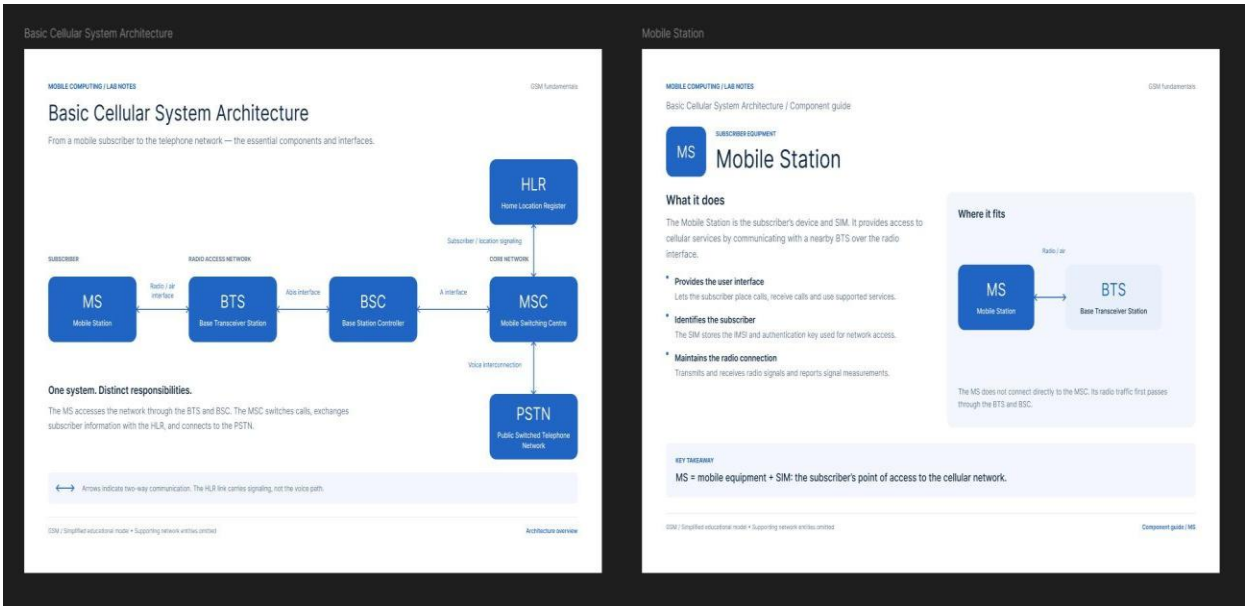
Aim:

To create an interactive diagram illustrating the components of a basic cellular system

Procedure: -

1. open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
11. Review and edit the Prototype
12. Save and Share

Design: -



Base Transceiver Station

MOBILE COMPUTING / LAB NOTES

GSM fundamentals

Basic Cellular System Architecture / Component guide

BTS

RADIO ACCESS NETWORK

Base Transceiver Station

What it does

The BTS contains the radio equipment that serves a cell. It handles the radio link with mobile stations and carries their traffic and signaling to the BSC.

- **Transmits and receives radio signals**
  Uses antennas and transceivers to communicate with mobile stations.
- **Processes the air interface**
  Performs radio modulation, channel coding and related signal processing.
- **Reports radio measurements**
  Supplies link-quality information for BSC-managed radio control.

Where it fits



The BTS is the radio endpoint of the network. The BSC coordinates its radio resources and handovers.

KEY TAKEAWAY

The BTS provides cell coverage; it is radio equipment, not the call-switching centre.

GSM / Simplified educational model • Supporting network entities omitted

Component guide / BTS

Base Station Controller

MOBILE COMPUTING / LAB NOTES

GSM fundamentals

Basic Cellular System Architecture / Component guide

BSC

RADIO ACCESS NETWORK

Base Station Controller

What it does

The BSC manages radio resources for a group of BTSs. It coordinates the radio access network and links it to the MSC for call switching and mobility services.

- **Allocates radio channels**
  Assigns and releases channels as subscribers start or end sessions.
- **Coordinates handovers**
  Controls handovers between BTSs under its control; other cases involve the MSC.
- **Controls radio operation**
  Manages radio power and resource use based on measurement reports.

Where it fits



The BTS–BSC connection uses Abis. The BSC–MSC connection uses the A interface.

KEY TAKEAWAY

The BSC manages the radio network; the MSC manages switching in the core network.

GSM / Simplified educational model • Supporting network entities omitted

Component guide / BSC

Mobile Switching Centre

MOBILE COMPUTING / LAB NOTES

GSM fundamentals

Basic Cellular System Architecture / Component guide

MSC

CORE NETWORK

Mobile Switching Centre

What it does

The MSC is the main switching node for circuit-switched mobile services. It routes calls and coordinates mobility and subscriber-related signaling with network databases.

- **Establishes and routes calls**
  Sets up, switches and releases circuit-switched call connections.
- **Coordinates mobility services**
  Works with location databases and manages handovers involving multiple BSCs.
- **Connects to other networks**
  Provides call interconnection toward external networks, often through a gateway MSC.

Where it fits



The MSC also connects to the BSC over the A interface. HLR signaling is separate from the voice connection to the PSTN.

KEY TAKEAWAY

The MSC switches calls and coordinates mobility; it does not provide the radio coverage.

GSM / Simplified educational model • Supporting network entities omitted

Component guide / MSC

Home Location Register

MOBILE COMPUTING / LAB NOTES

GSM fundamentals

Basic Cellular System Architecture / Component guide

HLR

CORE NETWORK / SUBSCRIBER DATABASE

Home Location Register

What it does

The HLR is the home network's database of subscriber records. It stores subscription information and a reference to the subscriber's current serving MSC/VLR.

- **Stores subscriber profiles**
  Maintains subscriber identities and the services permitted by the subscription.
- **Tracks the serving network**
  Updates the serving MSC/VLR reference when the subscriber registers in a new area.
- **Supports incoming-call routing**
  Helps the network obtain routing information for a mobile subscriber.

Where it fits



This simplified link represents subscriber and location signaling. Detailed GSM procedures also involve the VLR and gateway MSC.

KEY TAKEAWAY

The HLR stores subscriber and serving-location information; voice traffic does not pass through it.

GSM / Simplified educational model • Supporting network entities omitted

Component guide / HLR

Public Switched Telephone Network

MOBILE COMPUTING / LAB NOTES

GSM fundamentals

Basic Cellular System Architecture / Component guide

PSTN

EXTERNAL TELEPHONE NETWORK

Public Switched Telephone Network

What it does

The PSTN is the public circuit-switched telephone network. It enables mobile subscribers to exchange voice calls with fixed-line telephone users through network interconnection.

- **Serves fixed-line telephones**
  Connects telephone users through local exchanges and trunk networks.
- **Routes public telephone calls**
  Uses telephone numbers and switching systems to reach the called party.
- **Interconnects with the mobile core**
  Exchanges call signaling and voice traffic with the mobile switching network.

Where it fits



The diagram shows a simplified MSC–PSTN link. A gateway MSC commonly provides the boundary to the external telephone network.

KEY TAKEAWAY

The PSTN extends a mobile call beyond the cellular network to the public telephone system.

GSM / Simplified educational model • Supporting network entities omitted

Component guide / PSTN

Result: -

Hence an interactive diagram illustrating the components of a basic cellular system using Figma is created and executed successfully

2. Design a user interface prototype for a mobile station using Figma, considering connectivity and convenience.

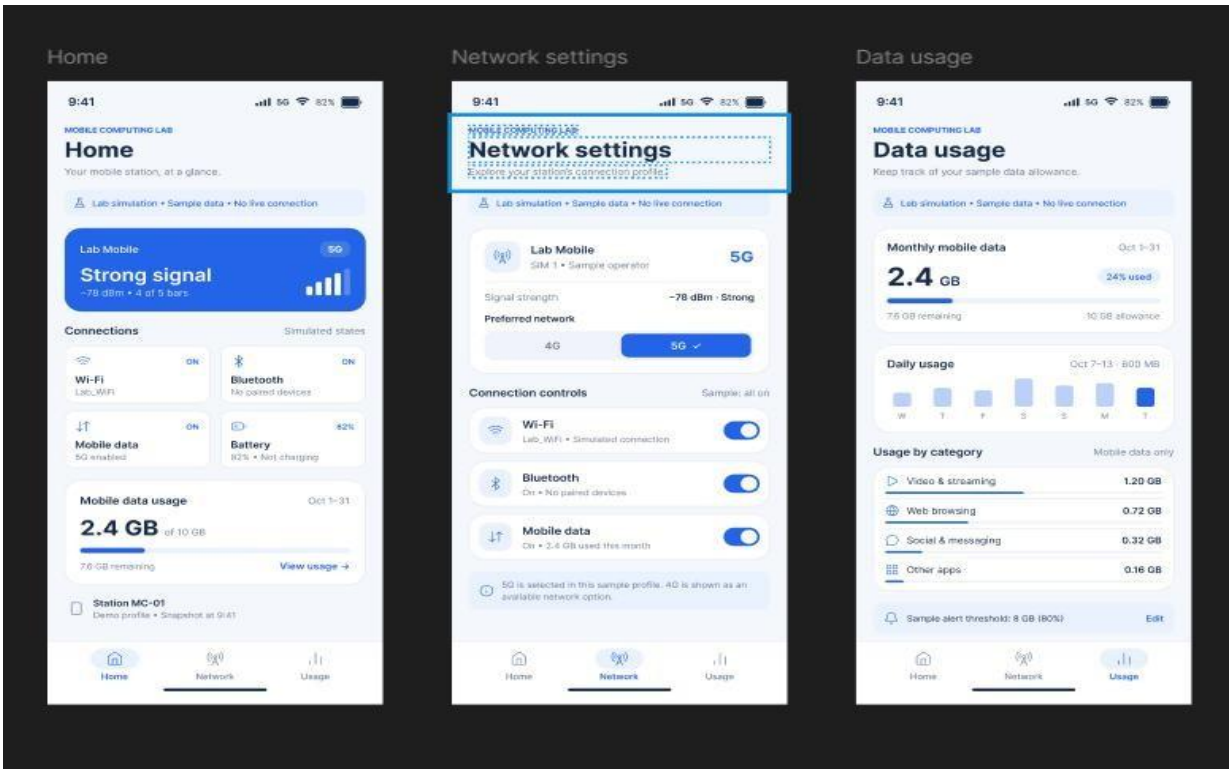
Aim: -

To create a user interface prototype for a mobile station using Figma, considering connectivity and convenience.

Procedure: -

1. open Figma
2. Create a new file
3. Select the Frames
4. Design Visual Elements
5. Make it Interactive
6. Add icons on the Frame
7. Incorporate Multimedia
8. Storyboard Animation
9. Review and edit the Prototype
10. Save and Share

Design:-



Result: -

A user interface prototype for a mobile station using Figma, considering connectivity and convenience using Figma is created and executed successful

3. Develop a flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system.

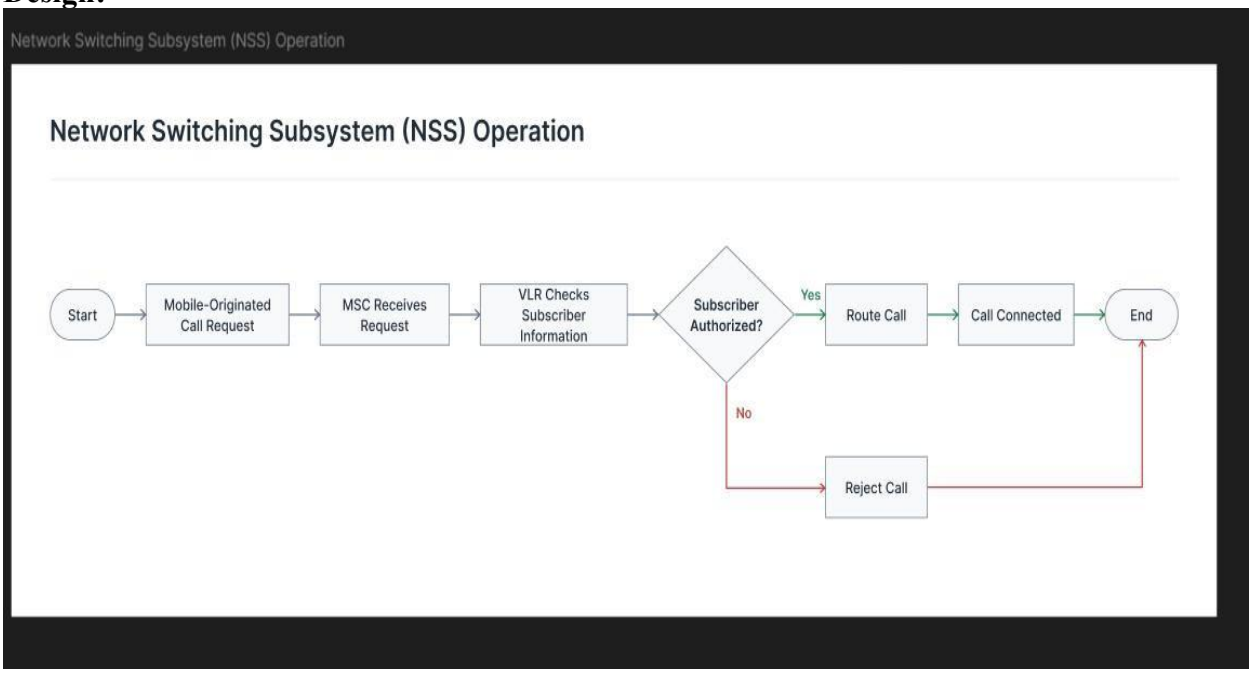
Aim: -

To draw a flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system

Procedure: -

1. open Figma
2. Create a new file
3. Click on the resources icon and click on the plugins
4. Search for the Flow Chart
5. Drag and drop the shapes on the screen
6. Now, enter the content on the shapes and connect them through Arrows.
7. Grouping and alignment
8. Review and edit
9. Save and share

Design: -



Result: -

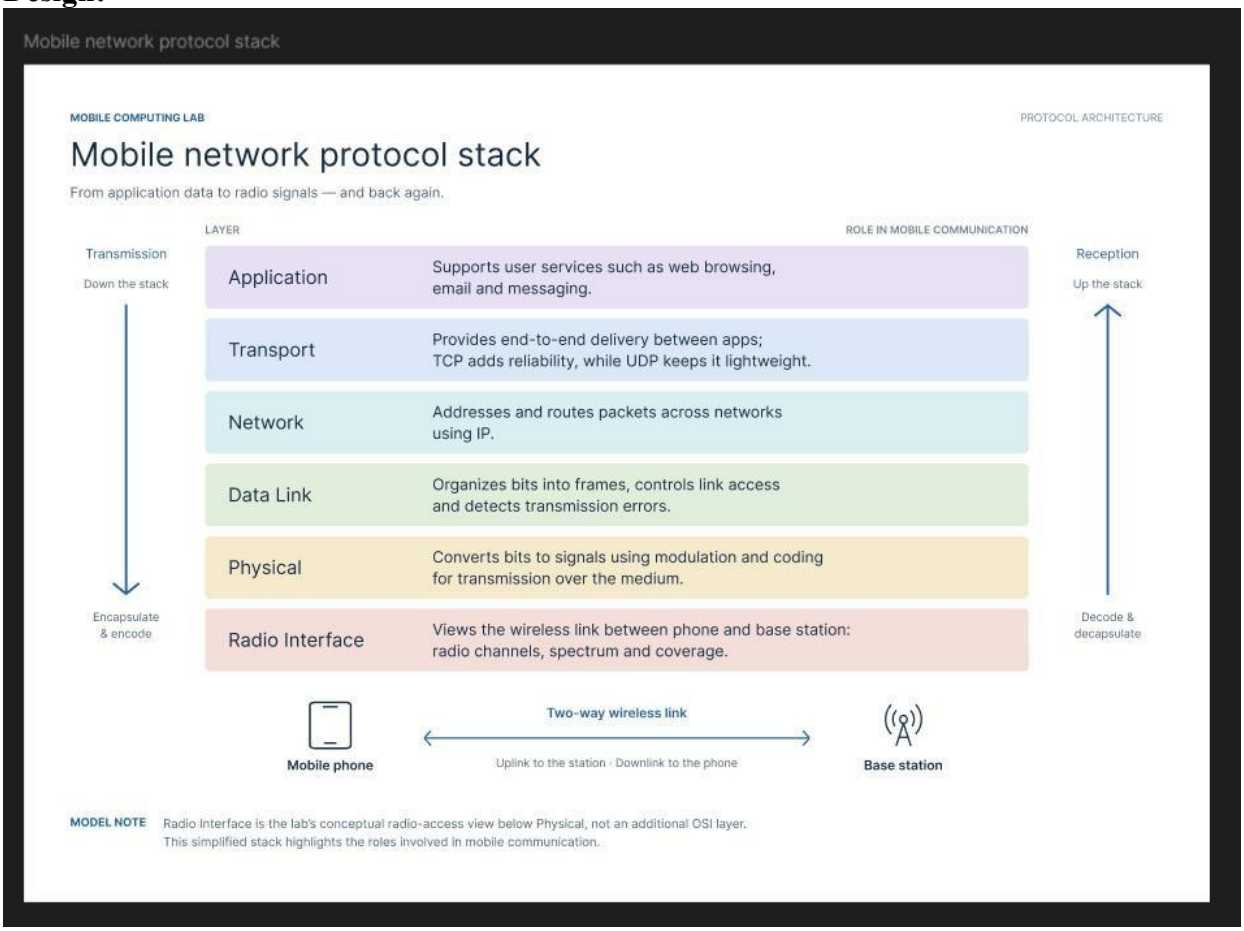
Hence the flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system has been drawn successfully.

4. Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Aim: -
Design a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Procedure: -
1. open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
11. Review and edit the Prototype
12. Save and Share

Design: -



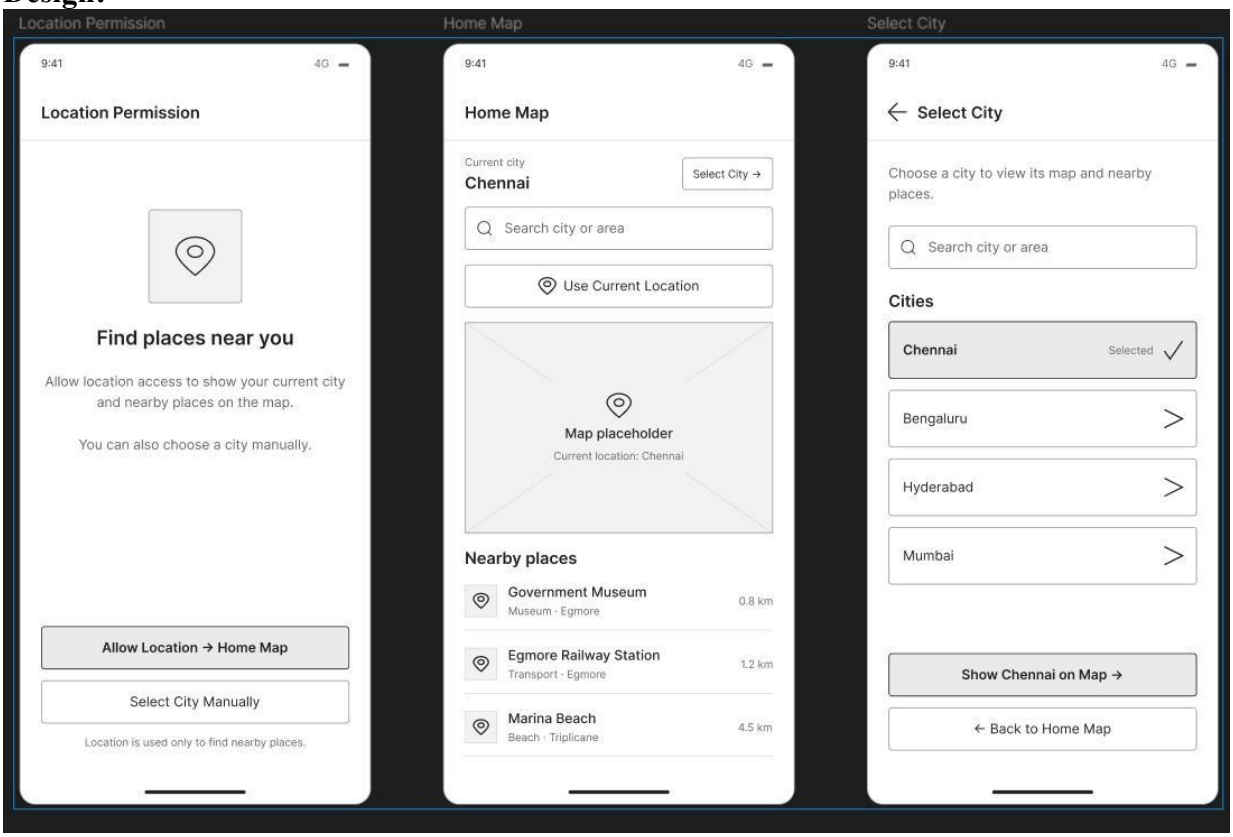
Result: -
Hence the visual representation in Figma highlighting the key elements of mobile network protocol stacks is created and executed successfully

5. Design a wireframe for a mobile application that incorporates localization features using Figma.

Aim: -
To design a wireframe using Figma for a mobile application that incorporates localization features

Procedure: -
1. open Figma
2. Create a new file
3. Select the Frames
4. Design Visual Elements
5. Make it Interactive
6. Add icons on the Frame
7. Incorporate Multimedia
8. Storyboard Animation
9. Review and edit the Prototype
10. Save and Share

Design: -



Result: -
Hence the wireframe for a mobile application that incorporates localization features using Figma is created and executed successfully

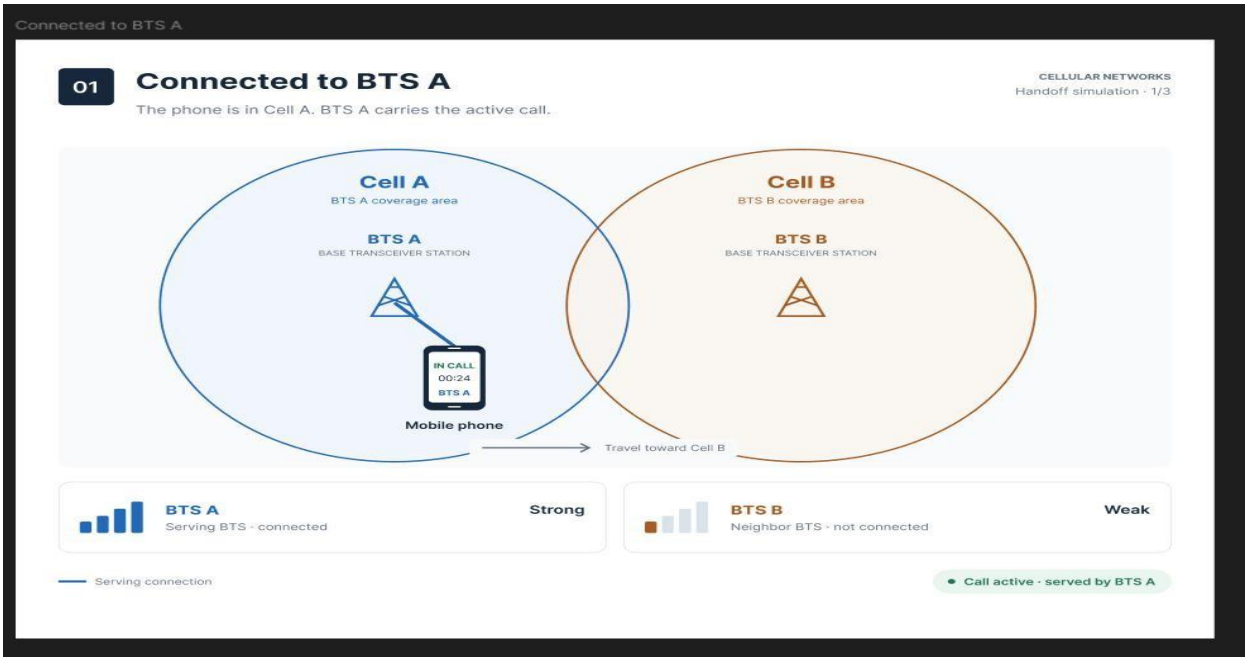
6. Use Figma to simulate a handoff scenario in a mobile network and visualize the process.

Aim: -To simulate a handoff scenario in a mobile network and visualize the process using Figma.

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
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10. Review and edit the Prototype
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Design:



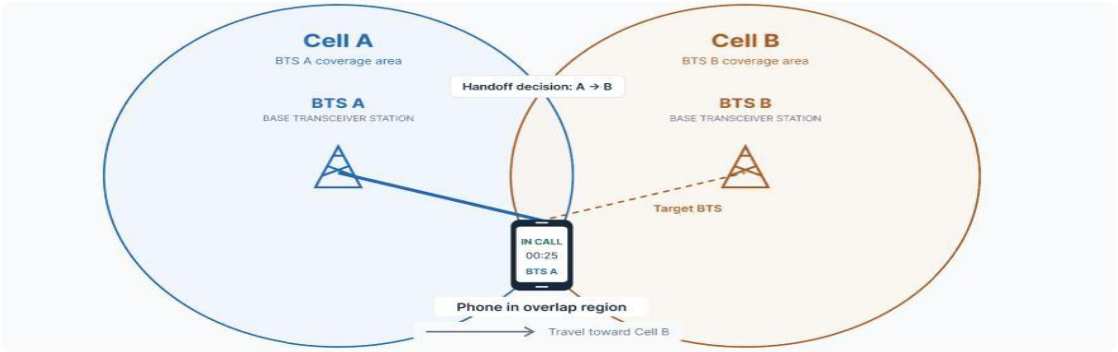
Moving toward Cell B

02

Moving toward Cell B

BTS A's signal decreases as BTS B's signal becomes stronger.

CELLULAR NETWORKS
Handoff simulation · 2/3



BTS A — Serving BTS · signal weakening — Decreasing ↓

BTS B — Target BTS · stronger than BTS A — Increasing ↑

Serving connection · Dashed = candidate link

• Call active · handoff to BTS B begins

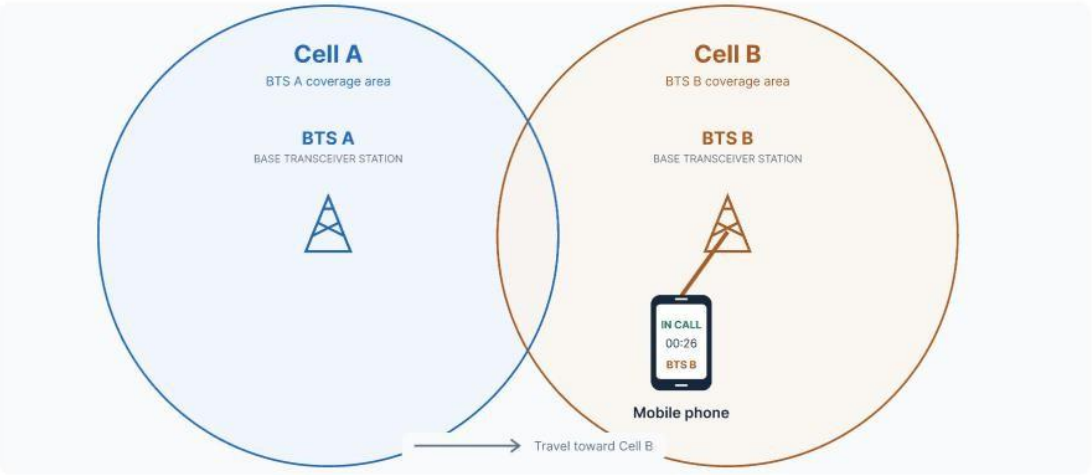
Connected to BTS B

03

Connected to BTS B

The handoff is complete. BTS B carries the same ongoing call.

CELLULAR NETWORKS
Handoff simulation · 3/3



BTS A — Previous BTS · no longer serving — Weak

BTS B — Serving BTS · connected — Strong

Serving connection

• Call continues · no interruption

Result:

Hence the simulated a handoff scenario in a mobile network and visualized the process using Figma.

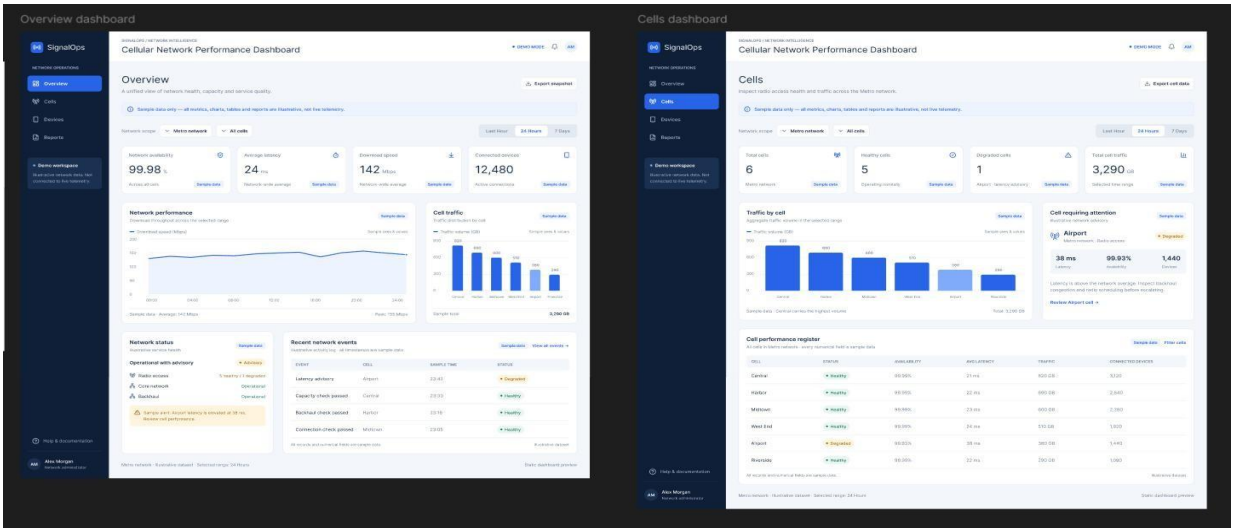
7. Develop a dashboard prototype in Figma displaying performance criteria for a cellular system.

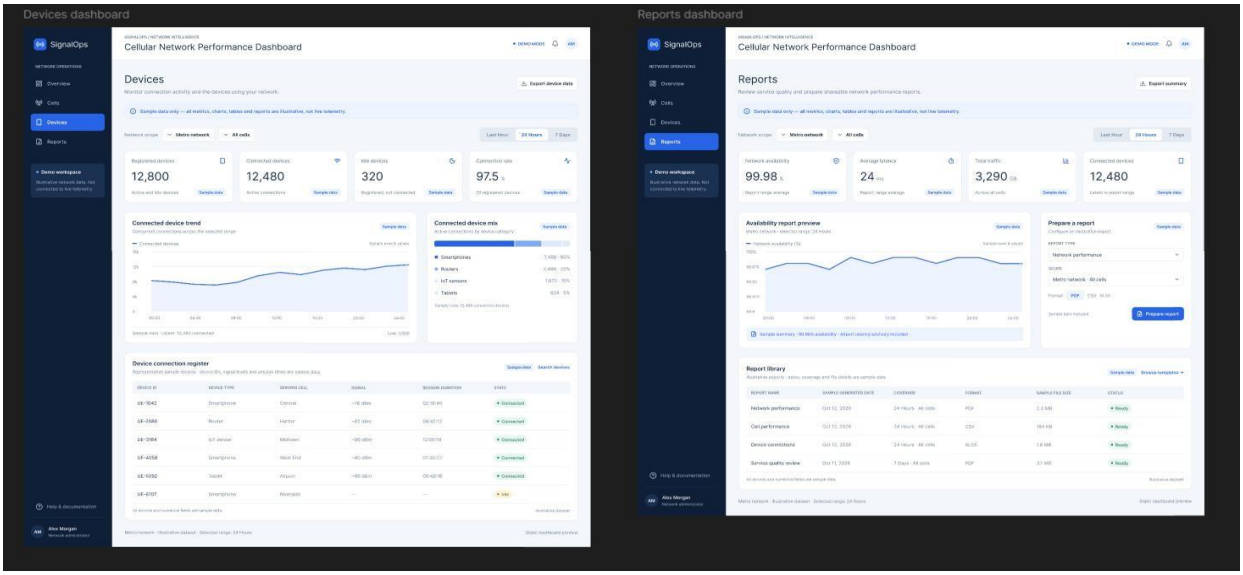
Aim: To develop a dashboard prototype in Figma displaying performance criteria for a cellular system.

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

Design:





Result:

Hence developed a dashboard prototype in Figma displaying performance criteria for a cellular system.

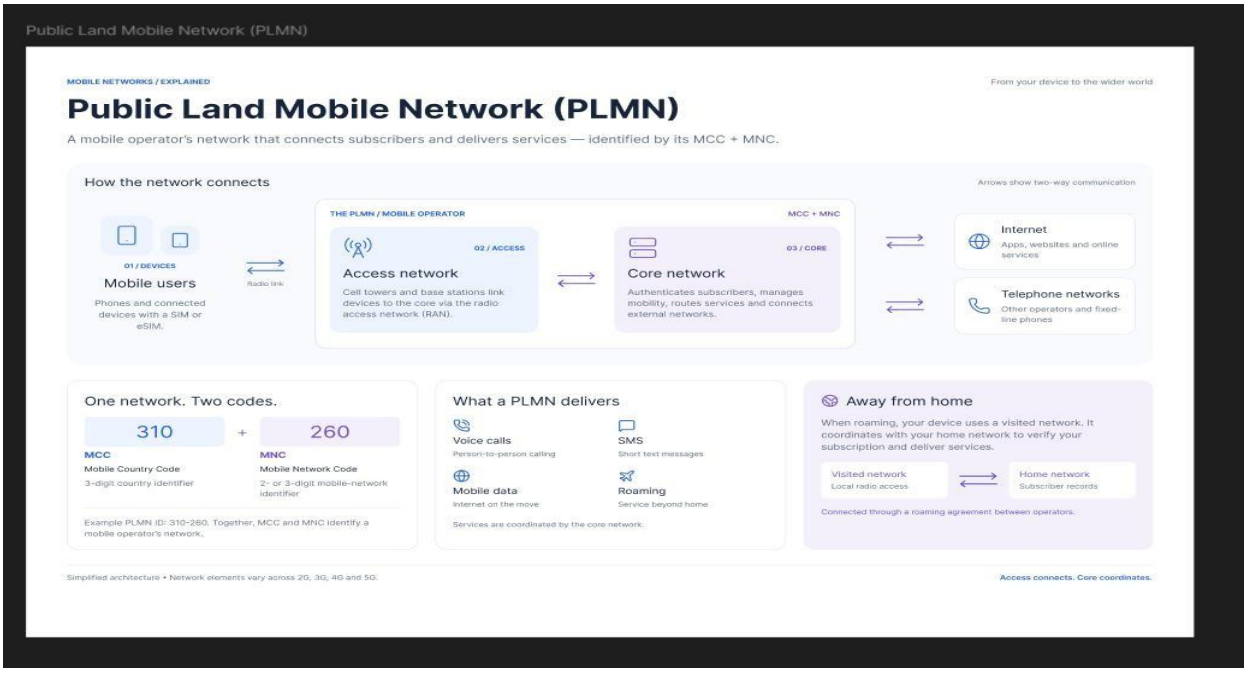
8. Create an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).

Aim: To create an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

Design:



Result:

Hence created an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).

9. Develop a user interface prototype for a GSM service using Figma.

Aim: To develop a user interface prototype for a GSM service using Figma.

Procedure:

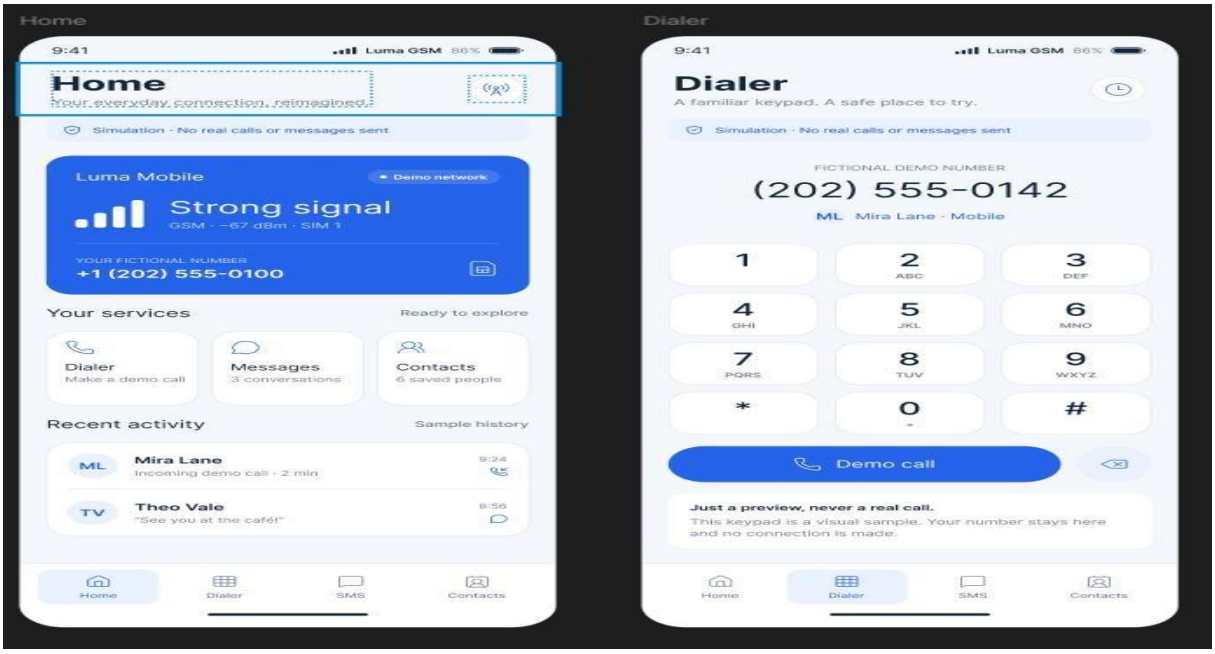
1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia

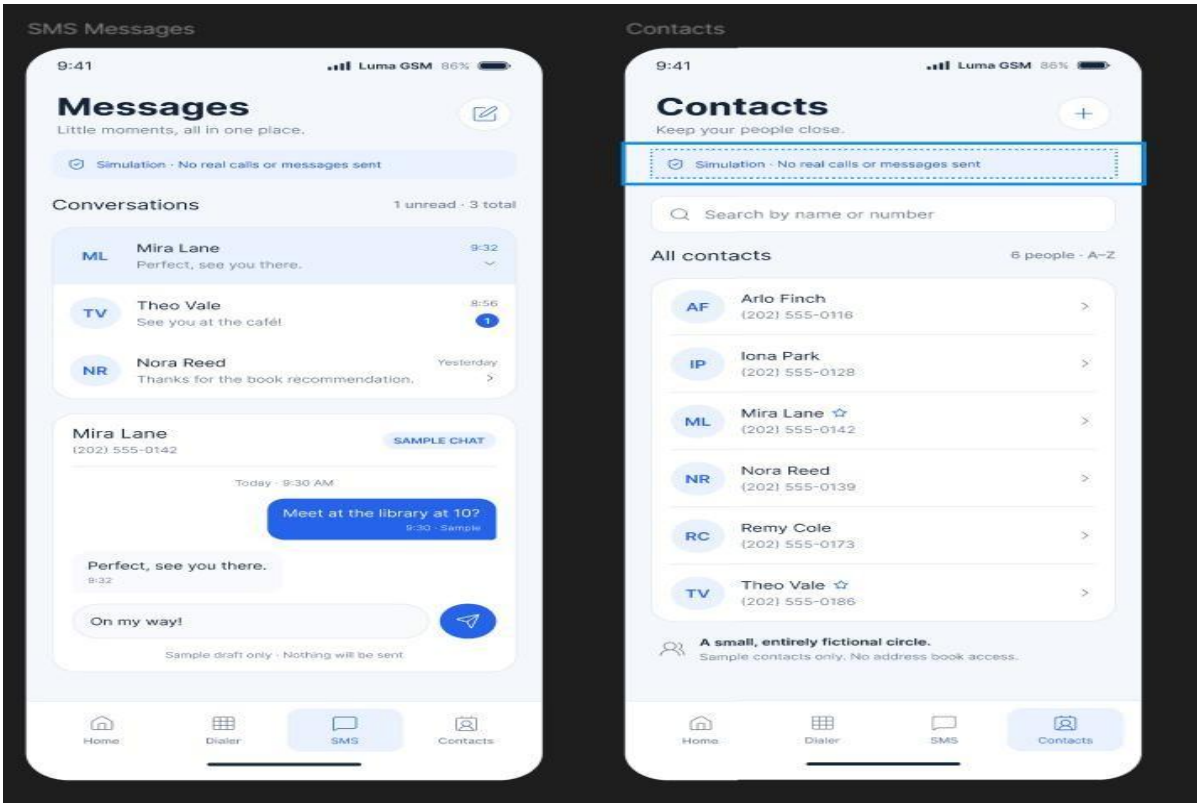
9. Storyboard Animation

10. Review and edit the Prototype

11. Save and Share

Design:





Result:

Hence developed a user interface prototype for a GSM service using Figma.

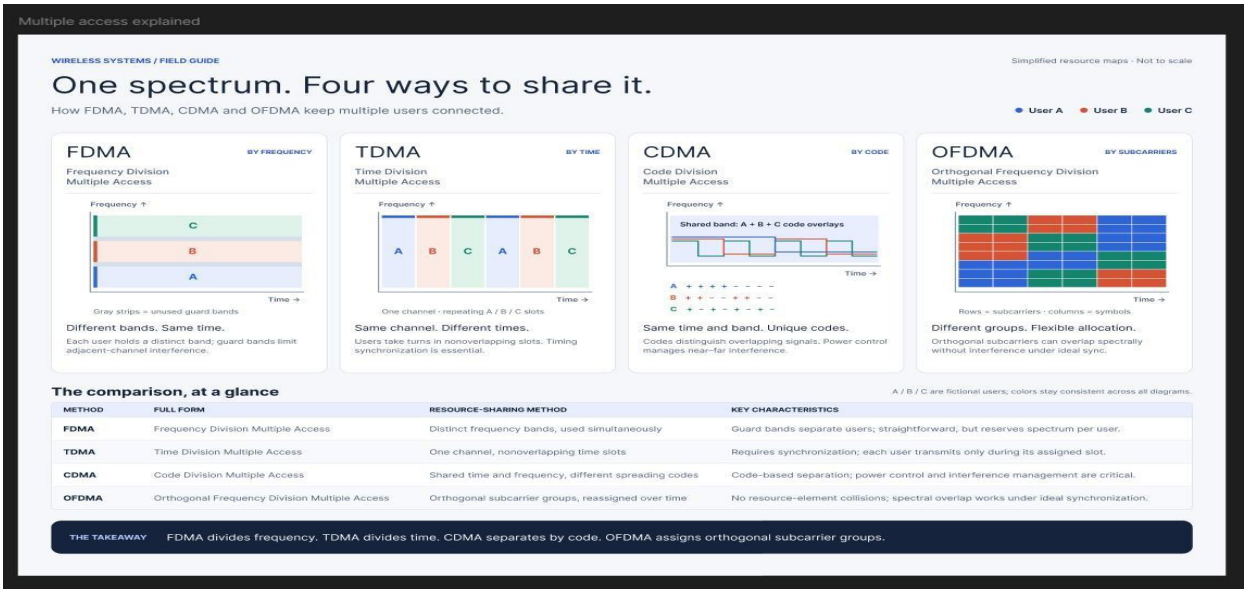
10. Create a visual representation in Figma showcasing the differences between various multiple access techniques.

Aim: To create a visual representation in Figma showcasing the differences between various multiple access techniques.

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
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Design:



Result:

Hence created a visual representation in Figma showcasing the differences between various multiple access techniques.